

ROBOTE: Interactive Educational Tool to Teach Basic Education Children to Classify and Collect Waste in their School Environment

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ABSTRACT

The presented project, denominated ROBOTE, aims to develop a robot which teaches children to classify waste. The objective of ROBOTE is to educate children of basic school age about the importance of correctly classifying waste and promoting sustainable habits in their daily life with a feedback-and-reward system to encourage participation and motivation. This project is rooted in the current problem of the increasing consumption of plastics and the lack of adequate waste management, and seeks to contribute to the achievement of Goals 12, 14 and 15 of the UN 2030 Agenda.

CCS CONCEPTS

• **Human-centered computing** → *Usability testing*; **Touch screens**; **User studies**; **Graphical user interfaces**.

KEYWORDS

Interface, children, waste, plastics, recycling, Design Thinking, usability testing, UN Goals

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1 INTRODUCTION

Waste is a major environmental problem worldwide. According to the World Bank report “*What a Waste 2.0: A Global Snapshot of Solid Waste Management to 2050*” [16], it is estimated that the amount of waste generated will reach 3.4 billion tons in 2050. It is important to classify waste to be able to handle it properly. Waste can be classified as organic, and inorganic, hazardous, and special. Recycling is one of the main ways to treat this waste and reduce its environmental impact.

Plastic waste pollution is of particular concern due to its durability and its ability to damage marine and terrestrial ecosystems. According to the United Nations Environment Program (UNEP), it

is estimated that around 8 million tons of plastic are dumped into the seas every year. This has led to the death of millions of marine fauna and the degradation of coastal and marine ecosystems.

To approach this problem, various projects and economic models have been developed. An example of this is the “*Circular Economy*” model, which seeks to maximize the use and recovery of resources and minimize waste and pollution. Policies and regulations have also been put in place, such as a ban on certain single-use plastic products to reduce the amount of plastic waste generated.

1.1 The UN 2030 Agenda and waste

The UN 2030 Agenda, also known as the Sustainable Development Goals (SDGs), is a global roadmap to eradicate poverty, protect the planet, and ensure that all people enjoy full and prosperous lives. One of the specific objectives of the 2030 Agenda is to end pollution, including pollution from plastic waste. Goal 14 “*Life Below Water*” seeks to reduce marine pollution caused by plastic waste, while Goal 15 “*Life on Land*” seeks to reduce land pollution caused by plastic waste. In addition, Goal 12 “*Sustainable Consumption and Production*” seeks to reduce the waste footprint and promote sustainable practices in the production and consumption of goods and services [20].

1.2 Plastic waste in Mexico and Oaxaca

In the case of Mexico, contamination with plastic waste is a serious problem due to the lack of infrastructure and adequate policies to manage waste. According to data from the National Institute of Ecology and Climate Change, Mexico is the ninth country in the world in the generation of plastic waste, with a recycling rate of 5% [7]. In addition, the problem is aggravated by the lack of awareness and education on proper waste management.

In Oaxaca (a state located in the South of the Mexican Republic), the problem is even more accentuated due to the lack of access to waste collection and treatment services in many rural communities and coastal towns. According to a study carried out by García-Sánchez [10], plastic waste on Oaxaca beaches represents 5% of total waste, with a greater presence in urban and tourist beaches, where a density of plastic waste of up to 1,350 pieces per km². In addition, the study also points out that most of the plastic waste found on the beaches of Oaxaca is of commercial origin, which suggests a lack of awareness and responsibility on the part of the producers and distributors of these products. Likewise, nowadays in Oaxaca there is a crisis in waste management, due to the overload in the collection of garbage generated and the lack of efficient policies to promote the classification of waste in homes. Government authorities and

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spaces designated for garbage collection are overwhelmed, which has generated an accumulation of garbage in the streets and an additional burden on the environment.

1.3 Education and awareness in children

Education and awareness are essential to address the problem of pollution by plastic waste. To encourage these practices in citizens, it is essential to establish waste classification routines. Adults, youth and children can help reduce consumption and encourage recycling. Children in particular can be sensitized to this issue through educational programs and practical activities, such as garbage collection in their community. In addition, children can set an example for their families and friends by promoting sustainable habits for caring for the environment. According to the National Pedagogical University [17], by teaching children to become responsible and productive members of society, significant and tangible benefits are more likely to be obtained from environmental education. Teaching-learning about environmental education serves as a fundamental tool to counteract environmental problems, and at the same time achieves the objective of environmental protection and conservation.

To assist in environmental education, books and other materials have been developed for use in the classroom. Several studies [3] have shown that there is a wide variety of educational materials available, although it is not enough, since sustainable practices still do not permeate most of society. To develop this type of material for schools, multimedia content is encouraged as valuable tools to improve education in schools, particularly for early-age students with poor reading skills, because it helps to visualize and understand complex concepts in relation to the collection and classification of plastic in a clearer and more appealing manner.

According to Hersh [11], he mentions that the use of multimedia in education can be useful to engage students in collaborative activities and encourage active learning, improving the accessibility of educational content for students with visual or hearing disabilities.

1.4 Recycling in Public Schools

Recycling in public schools is an important initiative to promote environmental awareness and reduce the amount of waste generated in educational institutions. Worldwide, many schools have implemented recycling programs to teach students about the importance of taking care of the environment and reducing environmental impact.

In Mexico and Oaxaca, recycling in public schools is an initiative that has been implemented in some parts of the country, but there is still a lot of room for improvement.

2 RELATED WORK

In Barar et al. [2] and Makhseed et al. [14] they propose low-cost smart containers that use the cloud and artificial intelligence to help sort waste. The former has 93.4% accuracy and low power consumption, while the latter focuses on metal, glass, and paper sorting. On the other hand, Ahmad's [1] project presents an interactive comic for preschool children that seeks to educate them about sustainable habits and environmental awareness through attractive stories and the use of multimedia. The results revealed that more than 70% of the participants found it easy to use and learn. The use

of multimedia in education has been shown to be important due to its positive impact on the teaching and learning process.

3 METHODOLOGY

The methodology selected for this project is Design Thinking. According to its authors, Tim Brown et al. [4], it is a problem-solving process and a methodology for innovation. He describes it as "iterative and user-focused, involving empathy, experimentation, and collaboration." This methodology allows the development of products and services that meet the needs of users effectively, by promoting collaboration between different disciplines and roles, which provides diversity in perspectives and solutions (see Figure 1), below.

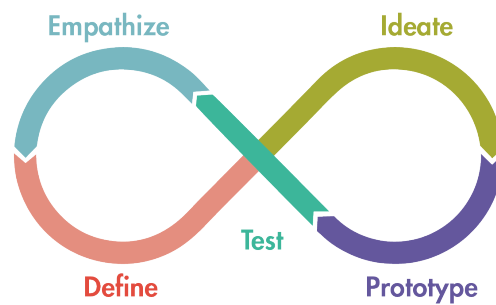


Figure 1: Design Thinking cycle and its focal points.

3.1 Initial idea

We started with a brainstorming session in which the current theme of the Student Design Competition was discussed along with the advisers. From this, three themes were considered, of which two were then merged (recycling and children). A bibliographic review and field research yielded the basic ideas to guide our project.

Next, we will present the implementation of Design Thinking through its five stages.

3.2 Empathize

According to Donald Norman [19] Empathize is stage which consists of understanding and feeling the needs, motivations and concerns of users. To carry out this stage, a Contextual Study technique was used, which is the process of collecting data about the users' context and thus obtaining a deep understanding of them [9].

The team conducted a contextual study at the "General Valerio Trujano" Federal Elementary School, in Huajuapán de León, Oaxaca, Mexico, to identify the problem of waste management in the school and to analyze behaviour patterns in children that contribute to this problem. Semi-structured interviews, surveys, observations, document analysis, records and data analysis were used to collect and analyze the information obtained (see Figure 2), below.

The study revealed that the waste that is most generated in the school is of organic origin, as well as plastic and paper, in addition to the fact that there is a lack of classification and collection



Figure 2: Contextual Study at the “General Valerio Trujano” Federal Elementary School, in Huajuapán de León, Oaxaca.

habits in the school. It was observed that basic education children are familiar with the use of technology, however, the school lacks specific programs and resources to develop skills and knowledge in this area. During the study, the workers responsible for cleaning the school (stakeholders) were also interviewed, whom reaffirmed the lack of sustainable habits in children and the little interest on the part of teachers in promoting environmental awareness.

Teachers commented that the COVID-19 pandemic had a significant impact on the basic education of children, causing a learning deficit, especially in subjects such as reading, writing and mathematics, left aside were issues related to sustainability, making it difficult for children to learn about caring for the environment from an early age.

From a Focus Group with operational personnel from SITRESUH (Integral System for Urban Solid Waste Treatment of Huajuapán - Responsible for waste collection in Huajuapán de León) and the CITRESO (Integral Center for Solid Waste Treatment - Responsible for the landfill sanitary of the municipality of Huajuapán de León), all our stakeholders as well, the following comments were obtained (see Figure 3):

- SITRESUH reported the collection from 45 to 50 tons of garbage, which are deposited in the CITRESO wastelands daily.
- The downtown area of the city generates the greatest amount of waste due to its commercial and residential activity.
- The authorities have launched various programs to encourage citizen participation in the correct classification of garbage, among them, a collection system by type of garbage and the “*Trueque para tu Bolsillo*” program that encourages the

exchange of PET or used cooking oil for coupons with monetary value, which can be exchanged for municipal services.

- In relation to schools, the possibility of institutions obtaining the garbage collection service for free was mentioned. However, it is required to be classified as organic waste, cardboard, PET and garbage in general. Unfortunately, most schools do not take advantage of this program due to the lack of interest and commitment of managers, cleaning staff, parents, and students to carry out actions that benefit their institution and their community.
- Likewise, SITRESUH has a specific program to collect organic waste generated in homes, businesses, and parks, which is used through various composting processes.



Figure 3: Focus Group with operational personnel from SITRESUH and CITRESO

3.3 Define

In this stage, objectives were established to address the needs of our users. One of the techniques applied was a Focus Group, that according to Krueger [13] it is a qualitative research technique in which a small team of people with similar characteristics meets to discuss a specific topic or product, guided by a moderator.

This activity was carried out in our UsaLab Laboratorio de Usabilidad at the Universidad Tecnológica de la Mixteca, with six basic education teachers from three different schools whom work with children from 6 to 12 years old, obtaining the following results (see Figure 4).

- Children learn complex topics more easily when they play collaborative and dynamic games with didactic material.
- Only 30% of the participants mentioned that recycling and waste collection activities were implemented at their school.
- 100% of the schools apply the “*tequio*” (A community practice of collective work in Oaxaca, Mexico. It is based on solidarity and mutual support among the members of the community. It is carried out voluntarily and without remuneration [6]). However, “*tequio*” only happens in a general way just a few times during the school year.
- All participants mentioned that environmental education must be accompanied by the collaboration of parents to



Figure 4: Focus Group with basic education teachers.

achieve a significant impact on awareness and the development of sustainable habits in children.

In the Focus Group, ideas were presented to solve the problem of waste management in schools, such as the use of attractive characters and interactive buttons to motivate children's participation and reward dynamics to encourage the desired behaviour. Participants suggested that the solution provide child-friendly data related to garbage sorting.

With the information obtained from the Contextual Study and the Focus Group, the necessary information was obtained to generate the user profile, through the Personas technique. It was defined that the project would be focused on children in a range of 6 to 7 years of age, who study in primary education schools, in the town of Huajuapán de León, Oaxaca, Mexico.

3.4 Ideate

According to the book *Design Thinking for Educators: Empathize, Define, Ideate, Prototype, Test.* by RJ. Brooke Reynolds [12], the Ideate stage seeks to generate a large number of possible solutions. According to Nigel Cross [5], this stage seeks to maximize creativity and encourage collaboration.

For this project, a Brainstorming dynamic was carried out to generate solutions. The team met in a collaborative environment with limited time to brainstorm and share ideas (see Figure 5). Tools such as *Mind Maps* were used to organize them. Subsequently, they were evaluated in relation to the following criteria: feasibility, social and economic impact, as well as cost. Taking into account the context of public schools in the state of Oaxaca.

As a result of this technique, the ROBOTE project was created. ROBOTE is an electronic device to guide children in the proper classification of garbage to make them aware of the importance of caring for the environment from an early age. Through multimedia content, ROBOTE shows the benefits of classifying correctly and promotes desirable environmental behaviors with the aim of achieving change in the short and long term, as well as promoting care for the environment. ROBOTE operating dynamics could be explained as follows:

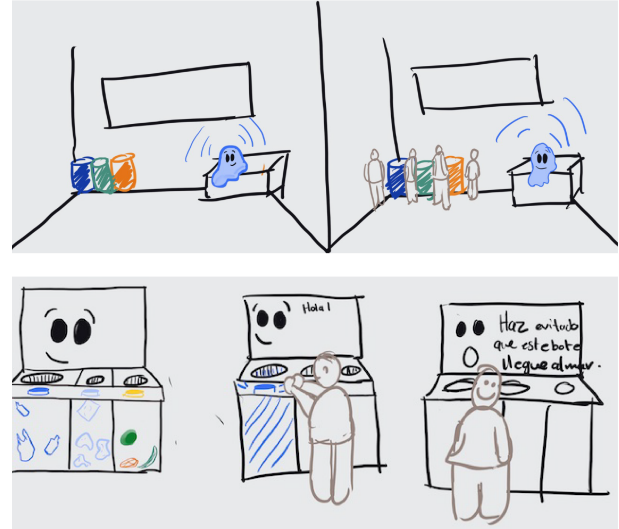


Figure 5: Sketching technique to illustrate the resulting ideas.

3.5 ROBOTE Operation

ROBOTE will be placed in the classroom of each participating group, and the teachers will provide the ROBOTE software with the names of the children in the group, who then will be assigned a badge with a QR code that will serve as identification before the system. The children will scan this code through the sensor placed at an anthropomorphically suitable height for them. From here, they will be able to access the system home interface.

In this interface, the Play button will be displayed, which will indicate to the users the beginning of the process. When the button is activated, they will access the main interface, where the sections are: Profile, Frequently Asked Questions and Classification. In the latter, 3 buttons will be shown representing the physical containers, the user must select the one that corresponds to the garbage he wishes to classify.

Once an option has been selected, the ROBOTE interface will show the user how to place the garbage into one of ROBOTE's three containers, then information will be presented indicating that the action was correct. Multimedia content will be shown that helps users visualize the scope of their actions when classifying, as well as the impact of not doing so, thus helping children to conceive of a realistic context. In addition, ROBOTE will give each child a badge when they reach a certain amount of classified garbage (organic, paper and/or plastic).

This badge system will be redesigned each school year, which will motivate the collection and interest throughout the academic life of children, as Doug Belshaw [8] mentions that badges can promote a culture of constant learning and the recognition of achievements throughout life. Mazar et al. [15] also point out that to achieve long-term environmental behavior, structural changes are required that encourage the repetition of desirable actions and discourage the habitual performance of those unwanted actions. In this sense, it is highlighted that repetition can make an action that was previously difficult become a habit, making it feel familiar, safe and easy to perform.

In accordance with the above, Verplanken [21] argues that establishing sustainable habits in daily life, such as reducing the use of plastic, recycling, or using public transport, can be an effective way to encourage more environmentally friendly behavior. Thus, the establishment of sustainable habits can lead to the automation of a positive behavior, facilitating its repetition and making it a habitual and natural practice.

In case of performing the action incorrectly, the system will offer information so that users can correct the action.

When users select the Profile section, they will be able to view their personal data and the amount of garbage they have classified so far. They will also be able to check the amount of trash left to get a badge, in addition to visualizing how many of them have won.

The Frequently Asked Questions section aims to guide the user in case of doubt, and sufficient information is offered to cover any doubts that may arise in the flow of actions that ROBOTE has. It also contains a guide for users who require support for any incident that would occur in the system (see Figure 6).

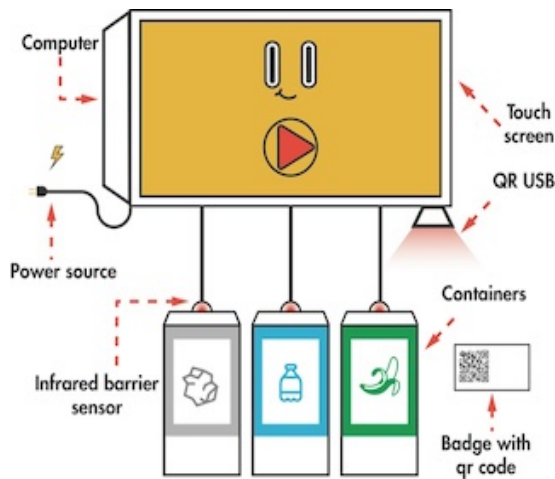


Figure 6: ROBOTE Operation

3.6 Technology used

ROBOTE is an electronic device that integrates a touch screen and a *Raspberry PI* computer, a low-cost and small size computer. It is based on an ARM processor and uses a version of a Linux operating system. It has wireless connectivity and storage capabilities. This computer has several input/output interfaces (GPIO) that allow us to connect sensors, actuators, and other external components. In addition, an infrared (IR) barrier sensor will be used on the bins to detect the entry of waste. For the detection of each child's credentials, a USB QR code reader is proposed, connected to the computer.

Apart from the functionalities of ROBOTE, the economic impact was also considered, therefore, the approximate cost of the construction per piece would be approximately \$95 USD, but if it is replicated in public schools throughout Mexico, the cost could be further reduced.

3.7 Prototype

In the Prototyping phase, the ideas generated were put into practice. To carry out this stage, a physical prototype was made on paper, which was evaluated in our UsaLab, where Wizard of Oz tests were carried out with the participation of five users, children from 6 to 7 years of age.

In the tests, the Thinking Aloud Protocol technique was used to assess three specific tasks. The results obtained in these tests allowed for the identification of indicators that were used to iterate on the design.

- The first task consisted in recognizing the elements present in the paper prototype. The participants clearly understood the icons that represented organic waste, paper, and plastics. However, they pointed out that the color white would be more suitable to represent paper waste.
- The second task consisted of depositing a plastic bottle in the appropriate container. When asked about the result of the task, they indicated that it had been correct, since the prototype indicated an expression of happiness.
- In the third and final test, the incorrect process was presented as what would happen if the sorting process was done incorrectly. The participants pointed out that the expression of sadness was adequate to indicate an erroneous process, but suggested that it would be useful to include an indication of how to correct the error.

In addition, the feedback received in this first evaluation indicated that the use of text could generate frustration in children who are learning to read. Therefore, it was decided to use large icons and buttons, as well as the implementation of audio that guides users in the flow of activities.

It is important to mention that parental permission was obtained by signing the document "Consent to Participation".

3.8 Test

The Test stage focuses on verifying and validating the solutions proposed in the project. In this case, usability tests were carried out with the ROBOTE prototype at the "Vicente Guerrero" Federal Elementary School, in Huajuapán de León, Oaxaca, with the aim of identifying possible usability improvements. According to Jakob Nielsen [18], at least five users are required to solve 80% of usability problems. In this study, ten users between the ages of 6 and 11 were invited to the tests.

To carry out the usability tests, the Thinking Aloud Protocol technique was used. In total, the user was asked to perform three specific tasks.

Users with specific tasks were evaluated. The results of these allowed us to conclude that the design is efficient and easy to use for users, and helps to identify areas of improvement for future developments.

For the usability tests, the user was contextualized in relation to the basic use of ROBOTE, providing enough information so that he could interact with the device himself.

- The first task consisted of the user accessing the ROBOTE system with their identification card (QR) through the scanner and placing the garbage where they considered appropriate. The participants mentioned that it is easy to use and

understand; however, one of the children performed the task incorrectly by choosing the wrong container, but was able to perform the task again with a positive result, verifying the ability to recover in case of an system error.

- For the second task, the user was asked to find the Questions section and return to the home screen. Most of the children had a positive result in recognizing the corresponding icon and correctly returning to the home screen.
- In task number three, the user had to check his Profile to check his progress level in order to get a badge and return to the home screen. All users recognized the profile section and understood the relationship between the progress shown by the system in classifying garbage correctly.

3.9 Redesign

For the third iteration of the design phase of the Design Thinking methodology, and with the results of the usability tests, provided elements to conceive the render of the high-fidelity prototype (see Figure 9). Once the objectives of functionality in relation to the needs of the user were achieved, the aesthetics were improved to create a visually attractive product through the use of colors and the arrangement of the elements. In addition, the specific anthropocentric measurements of Latin American children were considered[22], thus ensuring better adaptation and comfort for users. In other words, the prototype presents an ergonomic and functional design that allows easy and comfortable handling by children.

For the high fidelity prototype, the use of multimedia content that reinforces sustainable habits of children is proposed. For example, when a student deposits a PET bottle, ROBOTE will show on his screen the positive actions that result from that bottle not reaching municipal garbage dumps, water zones or city streets. ROBOTE will illustrate to children the processes of reusing materials such as PET, and how it is used again.

Likewise, when the child discards organic matter, ROBOTE will show how fertilizer or energy is generated from the garbage. By depositing paper, children will be able to visualize the process of making new paper. In this way, children will be able to understand in a fun and didactic way the importance of correctly collecting and classifying waste, and create habits that extend to the different contexts in which they develop, inviting their parents, siblings, relatives, and friends to carry out waste classification and collection actions.

4 RESULTS

The Design Thinking methodology was a key factor in the development of a viable prototype for sorting and collecting waste in their school setting.

For our overall project concept, we ran three iterations of the device in low (see Figure 7), medium (see Figure 8), and high fidelity (see Figure 9). The device only had changes in the design section, since the tasks were appropriate, but not the graphic elements.

For the first prototype, 15 tasks were implemented, of which 86% were successful, while for the second prototype, 30 tasks were carried out and had a success rate of 93%.

In order to evaluate the satisfaction of the device, a Likert scale measurement technique was used, which is used to measure the

attitude or opinion of a person towards a certain object, idea, or situation. In the case of our project, applying the Likert scale in children may be a bit different than in adults since it is important to take into account the children's reading and comprehension skills, so instead of using words to the response options were shown faces, representing from "strongly agree" to "strongly disagree".

Of the 10 children evaluated, 100% expressed acceptance and confidence in the tasks performed by the device.

For the third iteration, usability tests will be necessary with the high-fidelity prototype a real environment, which in this case will be the classroom, where users will be able to interact by classifying their waste.

It is also important to mention that for all the tests and studies Consent Letters were signed by the parents, teachers and principals of the various schools that participated in this project.



Figure 7: Low-fidelity prototype of ROBOTE



Figure 8: Medium-fidelity prototype of ROBOTE



Figure 9: High-fidelity prototype of ROBOTE

5 CONCLUSIONS

“Don’t underestimate the power of your small daily actions. Sorting and recycling your trash may seem like a small task, but it has a big impact on our planet and our community.”

After carrying out the tests, it was found that children between 6 and 7 years old were more enthusiastic and motivated in the activities and interaction with ROBOTE. The device is originally designed for classrooms in basic education schools; however, through iterations, it is possible to adapt it to place it in public spaces such as offices, shopping centers, or at home.

Through the development of ROBOTE, an interactive educational tool to teach basic education children to classify and collect waste in their school environment, it is possible to teach necessary habits for caring for the environment and contribute to solving the crisis of garbage dumps in Mexico. This robot uses technologies such as PIR sensors, QR code readers and Raspberry Pi to achieve efficient and motivating Human-Computer Interaction for children. In addition to this, by using a user-centered design approach, it was possible to create a device that was intuitive and easy to use for students.

In terms of the UN Sustainable Development Goals, this project contributes to the achievement of several of them, such as Goal 12, which seeks to ensure the sustainable management of natural resources; Goal 14, which seeks to conserve and sustainably use the oceans, seas, and marine resources for sustainable development, and Goal 15 which seeks to protect, restore, and promote the sustainable use of terrestrial ecosystems.

By educating children about the importance of properly classifying and collecting waste and by fostering a culture of caring for the environment, you are contributing to the creation of a more sustainable world for all. In addition, by using technologies such as the Raspberry Pi and sensors, the use of sustainable and efficient technologies is promoted, which contributes to the reduction of environmental impact and the efficient use of resources. In summary, the ROBOTE project is a valuable tool to improve the quality of education and promote a culture of caring for the environment in the next generations. At the same time, it contributes to the achievement of the UN sustainable development goals.

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